

Andrea Valente

Bologna, Italy

[linkedin.com/in/andrea-valente0](https://www.linkedin.com/in/andrea-valente0) · github.com/Valendrew

AI computer vision engineer who works on ill-defined problems, where defect criteria or success metrics have to be pinned down before building a pipeline that combines classical image processing, geometric fitting and deep learning on synthetic data. Works with clients on validation and next technical steps, and is moving towards roles that own that technical direction.

Experience

Deep Vision Consulting

Modena, Italy

AI Computer Vision Engineer

Apr 2024 - Present

Industrial package analysis, manufacturing R&D

- As primary implementation owner, built a Python analysis suite that assesses package samples from force signals and image sequences, which the client's R&D team used on new samples as an operational tool.
- Turned loosely specified defect criteria into measurable features: evaluated force-curve features, then added image segmentation and RANSAC geometric fitting where force alone could not separate defects from sample-preparation failures.
- With few real examples of a critical defect and no way to produce representative physical ones, synthesised defect geometry from annotated images and trained a U-Net for defect keypoints on the result.
- Designed and built the analysis dashboard end-to-end (Celery/RabbitMQ GPU job queue, PostgreSQL, cloud storage, Docker) for training, evaluation and inspection of failed samples; client engineers deployed it.

3D face reconstruction and measurement, global luxury group

- With no reconstruction KPIs, real-data benchmark or tolerances at the start, built the client's first quantitative evaluation framework for monocular 3D face reconstruction: registered annotated face scans to the topology of a 3D morphable head model and defined region-level error metrics and task-specific measurement KPIs.
- Built synthetic data and training pipelines for monocular reconstruction networks based on a 3D morphable head model, rendering with PyTorch3D and Blender and tracking experiments in MLflow.
- Designed the acquisition protocol for a multi-device capture study, then validated, processed and analysed the collected data.
- Built a reusable interactive analysis tool that breaks accuracy and repeatability down by device, head pose, distance and lighting; its results set the next technical priorities agreed with the client.

Efficient deep learning models for mobile inference, global luxury group

- Implemented the Python training, evaluation, export and tracking pipelines for a compact 6DoF head-pose model for browser virtual try-on; screened lightweight CNN backbones and built the comparison tooling behind the model the client approved as default.
- Benchmarked ONNX Runtime WebGPU against the TFLite baseline in the browser, counting GPU synchronisation and output retrieval in the timings, and integrated the WebGPU path with graph capture as the intended runtime for the pre-production application.

Point-cloud fitting, smart-device manufacturer

- Developed a Python package for RANSAC-based fitting of geometric primitives to point clouds of smart-lighting installations, with synthetic point-cloud generators configurable for noise, outliers and missing faces to test the fitting.

AI Computer Vision Engineer (Intern)

Sep 2023 - Mar 2024

- Master's thesis on 3D face reconstruction for virtual try-on: evaluated NeRF/MVS against monocular 3D morphable-model methods for a few-seconds, single-image target, continued with morphable-model inference, and trained a face-shape regressor on synthetic renders.

Vaia Innovations

Remote, part-time

Frontend Web Developer

Jun 2022 - Sep 2022

- Built responsive interfaces from Figma designs for article exploration and writing on a Web3 publishing platform, using TypeScript, Vue.js and Tailwind CSS.

Skills

Computer vision: OpenCV, scikit-image, U-Net segmentation and keypoint detection, RANSAC, 3D morphable models, mesh registration, point-cloud processing, Open3D

Machine learning: PyTorch, synthetic data (Blender, PyTorch3D), MLflow, model evaluation, ONNX Runtime, WebGPU, TFLite

Engineering: Python, NumPy, Polars, Docker, Celery, RabbitMQ, PostgreSQL, Playwright, Git, TypeScript, Vue.js

Projects

Comparative argument retrieval and stance detection

GitHub · MSc, 2022/23

Three-person team project comparing sparse, dense, hybrid and MonoT5-reranked retrieval for comparative questions, with DistilBERT-based stance classification.

POS tagging and abstractive question answering

GitHub · MSc, 2022/23

Team comparisons of recurrent POS taggers (BiLSTM, BiGRU) and transformer encoder-decoder question-answering models (TinyBERT, DistilRoBERTa).

Crime-category classification on Hadoop/Spark

GitHub · MSc, 2022/23

PySpark preprocessing, feature design and classification on a three-node cluster, using only information available at prediction time.

Rectangular circuit placement

GitHub · MSc, 2021/22

Two-person project on non-overlapping placement with minimum plate height across CP, SMT and MIP, focusing on the constraint-programming models and Python solver comparisons.

Counterfactual explanations

GitHub · MSc

Co-developed a comparison of optimisation-based and DiCE counterfactuals for smartphone price prediction.

Education

MSc in Artificial Intelligence, University of Bologna

Sep 2021 - Mar 2024

110/110 with honours

Thesis: A study of 3D face reconstruction methods: from geometry to deformable mesh models.

Selected coursework: Machine Learning and Deep Learning; Machine Learning for Computer Vision; Image Processing and Computer Vision; Natural Language Processing; Combinatorial Decision Making and Optimization.

BSc in Computer Science, University of Bologna

Sep 2018 - Dec 2021

110/110 with honours

Thesis: DeblurGAN-v2 for motion-blur removal in tomographic images, compared against SelfDeblur.

ELLIS Winter School on Foundation Models (FoMo), Amsterdam

Mar 2026

Certifications

ICVSS 2025, International Computer Vision Summer School: exam certificate

Jul 2025

Languages

Italian: native. English: professional working fluency; MSc taught in English; IELTS Academic 7.0 (2021).